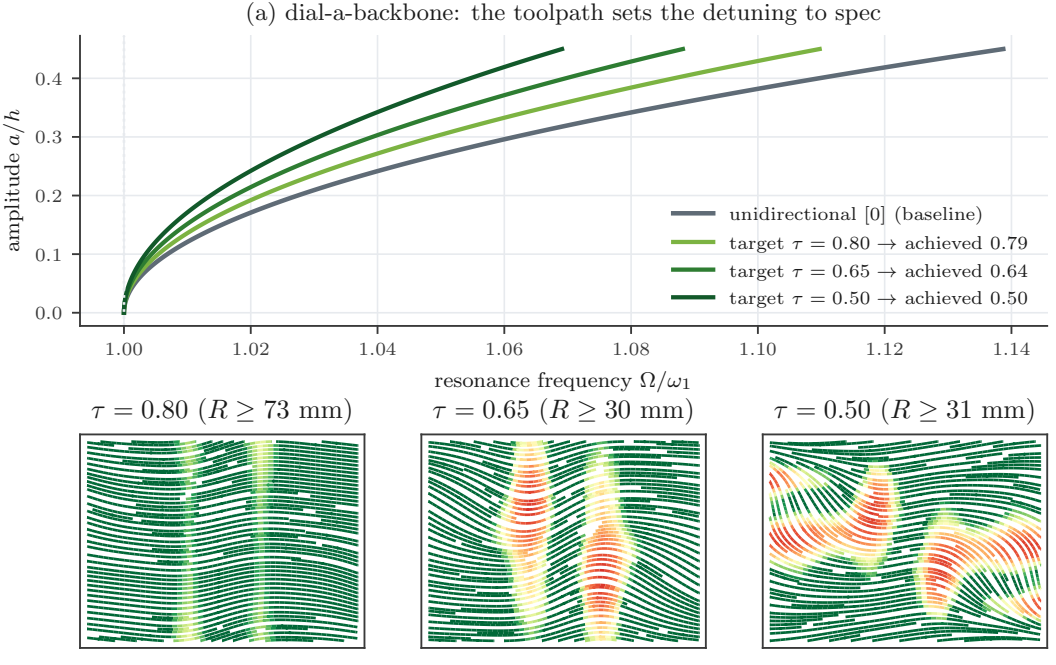

(a) dial-a-backbone: the toolpath sets the detuning to spec
amplitude a/h
0.0
0.1
0.2
0.3
0.4
1.00
1.02
1.04
1.06
1.08
1.10
1.12
1.14
resonance frequency Ω/ω_1
unidirectional [0] (baseline)
target $\tau = 0.80 \rightarrow$ achieved 0.79
target $\tau = 0.65 \rightarrow$ achieved 0.64
target $\tau = 0.50 \rightarrow$ achieved 0.50
$\tau = 0.80$ ($R \geq 73$ mm)
$\tau = 0.65$ ($R \geq 30$ mm)
$\tau = 0.50$ ($R \geq 31$ mm)